
**TEST OF MATHEMATICS FOR
UNIVERSITY ADMISSION****TMUA/02****Challenge Paper 2****75 minutes****Mathematical Reasoning****INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but do not open this question paper until you are told that you may do so.

A separate answer sheet is provided. Please check you have one. You also require a soft pencil and an eraser.

This paper contains twenty multiple-choice questions and should be attempted in 75 minutes. Each question has between six and ten options, labelled from A, exactly one of which is correct.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark. Record your choice for each question on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

You can use the question paper for rough working, but no extra paper is allowed. Only your responses on the answer sheet will be marked.

Calculators and dictionaries may NOT be used.

This is a Challenge Paper. It is set deliberately above the standard of the live test: the questions reward candidates who can combine several techniques and resist the obvious trap. Do not be discouraged by a low score.

Please wait to be told you may begin before turning this page.

This paper consists of twenty questions, a marked answer key, and full worked solutions.

1 Which one of the following statements is true?

- A** For every real number x , $\sqrt{x^2} = x$.
- B** The product of two irrational numbers is always irrational.
- C** If n is an integer and n^2 is even, then n is even.
- D** $0.\bar{9} < 1$.
- E** Every prime number is odd.
- F** $\log(a + b) = \log a + \log b$ for all positive reals a, b .

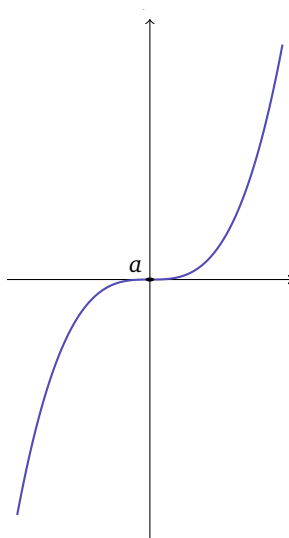
2 Consider the following two statements about an integer $n > 3$.

1. If n is prime, then $n = 6k + 1$ or $n = 6k - 1$ for some integer k .
2. If $n = 6k + 1$ or $n = 6k - 1$ for some integer k , then n is prime.

Which of them are true?

- A** Neither
- B** 1 only
- C** 2 only
- D** Both 1 and 2

- 3 The diagram shows part of the graph of $y = f(x)$ near a point $x = a$ where $f'(a) = 0$. Let P be the statement “ $f'(a) = 0$ ” and Q be “ f has a turning point at $x = a$ ”. Which option is correct?



- A P is neither necessary nor sufficient for Q
- B P is necessary but not sufficient for Q
- C P is sufficient but not necessary for Q
- D P is necessary and sufficient for Q

- 4 Consider the statement: “for every function f defined on $0 \leq x \leq 1$, the value $\int_0^1 f(x) dx$ equals the area enclosed by $y = f(x)$, the x -axis, $x = 0$ and $x = 1$.” Which of the following is a counterexample?
- A $f(x) = (x - 2)^2$
 - B $f(x) = x^2 + 1$
 - C $f(x) = 2x - 1$
 - D $f(x) = \cos x$
 - E $f(x) = e^x$
 - F $f(x) = 1 + \cos x$
 - G the statement is correct, so there is no counterexample

5 It is given that $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$. Here is an attempt to prove $0 = 1$.

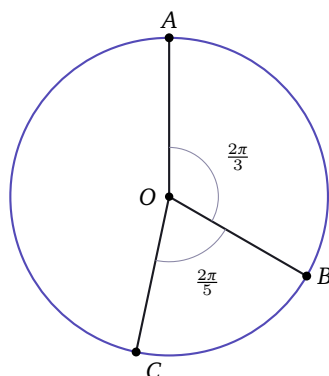
1. Let $u = \frac{1}{x}$ and $\frac{dv}{dx} = 1$.
2. Then $\frac{du}{dx} = -\frac{1}{x^2}$ and $v = x$.
3. So $\int \frac{1}{x} dx = \frac{1}{x} \cdot x - \int x \left(-\frac{1}{x^2}\right) dx = 1 + \int \frac{1}{x} dx$.
4. Subtracting $\int \frac{1}{x} dx$ from both sides gives $0 = 1$.

The proof is incorrect. Where is the first error?

- A** line 1
- B** line 2
- C** line 3
- D** line 4
- E** there is no error; the proof is correct

- 6** A repunit is a positive integer written using only the digit 1 (for example 1, 11, 1111). Complete the sentence: a repunit with n digits is divisible by 3 if and only if...
- A** n is a multiple of 3
 - B** n is a multiple of 9
 - C** n is a multiple of 7
 - D** n is even
 - E** it holds for every n
 - F** n is odd
 - G** it holds for no n

- 7 A , B and C are three vertices of a regular n -sided polygon inscribed in a circle with centre O . It is given that $\angle AOB = \frac{2\pi}{3}$ and $\angle BOC = \frac{2\pi}{5}$. Then n is necessarily a multiple of k . What is the largest value of k for which this must be true?



- A 15
- B 3
- C 5
- D 30
- E 8
- F 45
- G no such k exists

- 8 For how many of the following functions is f strictly increasing for all real x (on its whole domain)?

1. $f(x) = x^3$
2. $f(x) = e^x$
3. $f(x) = x + \sin x$
4. $f(x) = x^2$
5. $f(x) = 2x - \cos x$

A 0

B 1

C 2

D 3

E 4

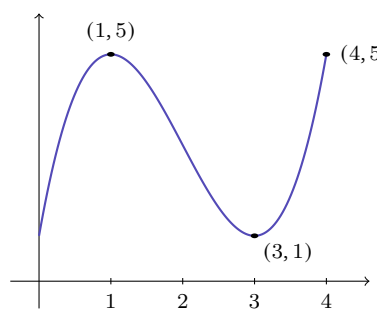
F 5

- 9** All 24 arrangements of the letters of the word MATH are listed in alphabetical order. In which position does the word MATH appear?
- A** 9
 - B** 12
 - C** 13
 - D** 14
 - E** 15
 - F** 18
 - G** 20
 - H** 24

- 10** Find the full range of values of the real number k for which $(\ln x)^2 - 4 \ln x + k = 0$ has exactly two distinct real solutions.
- A** $k < 4$
 - B** $k > 4$
 - C** $0 < k < 4$
 - D** $k < 0$
 - E** $k > 0$
 - F** $k \leq 4$
 - G** all real k

- 11** A squarefree integer is a positive integer not divisible by the square of any prime. Which of the following statements are correct?
1. A squarefree integer with exactly n distinct prime factors has exactly 2^n positive divisors.
 2. The product of two squarefree integers is always squarefree.
 3. The integer 1 is squarefree.
- A** None
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 12** The diagram shows $y = x^3 - 6x^2 + 9x + 1$ on the interval $0 \leq x \leq 4$. Find the maximum value of the function on this interval.



- A** 1
- B** 3
- C** 5
- D** 6
- E** 17
- F** 4
- G** -4

- 13** How many strings of 4 letters, each chosen from $\{A, B, C\}$, contain at least one A and at least one B ? (Order matters, and letters may repeat.)
- A** 14
 - B** 36
 - C** 50
 - D** 65
 - E** 81
 - F** 49
 - G** 32

14 Which of the following six numbers is the largest?

A $2^{3/2}$

B $3^{0.9}$

C e

D $\pi^{0.8}$

E $\ln 15$

F $5^{0.6}$

- 15** A function f is defined on all of $\mathbb{R}$, but is not assumed to be continuous. Let J be “ $f(1) = 2$ and $f(4) = -3$ ” and let K be “ $f(x) = 0$ for some x with $1 < x < 4$ ”. Which option is correct?
- A** J is sufficient for K
 - B** J is necessary for K
 - C** J is necessary and sufficient for K
 - D** J is neither necessary nor sufficient for K
 - E** K is sufficient for J
 - F** K is necessary for J

- 16** Complete the sentence: for a positive integer n , the congruence $n^2 \equiv 1 \pmod{24}$ holds if and only if...
- A** n is odd
 - B** n is prime
 - C** n is coprime to 6
 - D** n is not a multiple of 3
 - E** $n \equiv 1 \pmod{24}$
 - F** $n \equiv \pm 1 \pmod{24}$
 - G** it holds for every n

- 17** You wish to prove the statement “if n^2 is even then n is even”. Which one of the following arguments actually proves it?
- A** Assume n is even and show n^2 is even.
 - B** Assume n is odd and show n^2 is odd.
 - C** Check that $n = 2, 4, 6, 8$ all have n^2 even.
 - D** Assume n^2 is odd and show that n is even.
 - E** Assume n^2 is even and show that n^2 is a multiple of 4.

- 18** Three boxes are labelled “Apples”, “Oranges” and “Mixed”. You are told every label is wrong. You may draw one fruit from a single box of your choice and look at it. From which box should you draw to guarantee you can deduce the contents of all three boxes?
- A** the box labelled “Apples”
 - B** the box labelled “Oranges”
 - C** the box labelled “Mixed”
 - D** any single box will work
 - E** it cannot be done with a single draw

19 Which one of the following inequalities does not hold for all positive real numbers a , b , c ?

A $a^2 + b^2 \geq 2ab$

B $a + b \geq 2\sqrt{ab}$

C $a^2 + b^2 + c^2 \geq ab + bc + ca$

D $a^2 + b^2 + c^2 \geq 2(ab + bc + ca)$

E $(a + b + c)^2 \geq 3(ab + bc + ca)$

F $\frac{a}{b} + \frac{b}{a} \geq 2$

- 20** The real numbers x, y, z satisfy

$$x^2 + 2y = -1, \quad y^2 + 2z = -1, \quad z^2 + 2x = -1.$$

Find the value of $x + y + z$.

- A** -3
- B** 0
- C** 3
- D** -1
- E** 1
- F** there are no real solutions
- G** it cannot be determined

Answer key

Paper 2	
Q	Ans
1	C
2	B
3	B
4	C
5	D
6	A
7	A
8	E
9	D
10	A
11	F
12	C
13	C
14	A
15	D
16	C
17	B
18	C
19	D
20	A

Worked solutions

1 (C). A is false at $x = -1$ (there $\sqrt{x^2} = 1 \neq -1$); B is false, as $\sqrt{2} \cdot \sqrt{2} = 2$ is rational; D is false, since $0.\overline{9} = 1$ exactly; E is false (2 is prime); F is false in general. The contrapositive of “ n^2 even $\Rightarrow n$ even” is “ n odd $\Rightarrow n^2$ odd”, which is true, so C is the true statement.

2 (B). Every prime greater than 3 is coprime to 6, hence congruent to $\pm 1 \pmod{6}$: statement 1 is true. The converse fails, for example $25 = 6 \cdot 4 + 1$ is of the required form yet is not prime: statement 2 is false. So 1 only.

3 (B). At a turning point the derivative is zero, so $f'(a) = 0$ is necessary for Q. It is not sufficient: the diagram shows $y = x^3$, which has $f'(0) = 0$ but no turning point (the curve keeps increasing through an inflection). Hence P is necessary but not sufficient.

4 (C). The identity $\int_0^1 f = \text{area}$ fails whenever f dips below the x -axis, because area counts the region below the axis positively while the integral counts it negatively. Of the options, only $f(x) = 2x - 1$ is negative on part of $[0, 1]$ (namely $0 \leq x < \frac{1}{2}$); there $\int_0^1 (2x - 1) dx = 0$ but the enclosed area is $\frac{1}{2}$. The others stay non-negative throughout, so C is the counterexample.

5 (D). Every individual step of algebra is valid: line 3 is a correct integration by parts. The flaw is in line 4. An indefinite integral is defined only up to an additive constant, so the two copies of $\int \frac{1}{x} dx$ need not be the same number and cannot be cancelled. The apparent $0 = 1$ is really $0 = 1 + C$ for an arbitrary constant, which is no contradiction.

6 (A). A repunit with n digits has digit sum n , and an integer is divisible by 3 exactly when its digit sum is. So the repunit is divisible by 3 if and only if $3 \mid n$.

7 (A). A central angle between two vertices of a regular n -gon is a whole-number multiple of $\frac{2\pi}{n}$. From $\angle AOB = \frac{2\pi}{3} = a \cdot \frac{2\pi}{n}$ we get $n = 3a$, so $3 \mid n$; from $\angle BOC = \frac{2\pi}{5} = b \cdot \frac{2\pi}{n}$ we get $n = 5b$, so $5 \mid n$. Hence n is a multiple of $\text{lcm}(3, 5) = 15$, and 15 is the largest value forced.

8 (E). A function is strictly increasing when its derivative is non-negative and vanishes only at isolated points. Here x^3 ($f' = 3x^2$), e^x , $x + \sin x$ ($f' = 1 + \cos x \geq 0$) and $2x - \cos x$ ($f' = 2 + \sin x > 0$) are all strictly increasing; x^2 decreases for $x < 0$, so it is not. That is 4 of the five.

9 (D). In alphabetical order the letters rank A, H, M, T. Words beginning with A or H fill the first $2 \times 3! = 12$ places, so the M-words start at position 13. Among these, MA... comes first: MAHT is 13th and MATH is 14th.

10 (A). Put $t = \ln x$, which takes every real value exactly once as x ranges over $x > 0$. The quadratic $t^2 - 4t + k = 0$ has two distinct real roots precisely when its discriminant is positive: $16 - 4k > 0$, i.e. $k < 4$. Each root t gives one $x = e^t$, so there are two solutions if and only if $k < 4$.

11 (F). Statement 1 is true: a squarefree number $p_1 p_2 \cdots p_n$ has 2^n divisors. Statement 3 is true: 1 has no prime factors, so it is (vacuously) squarefree. Statement 2 is false: 6 and 10 are squarefree but $6 \times 10 = 60 = 2^2 \cdot 3 \cdot 5$ is not. So 1 and 3 only.

12 (C). $f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3)$, so the interior stationary points are $x = 1$ (local maximum) and $x = 3$ (local minimum). Evaluating at these and at the endpoints: $f(0) = 1$, $f(1) = 5$, $f(3) = 1$, $f(4) = 5$. The maximum on the interval is 5 (attained at both $x = 1$ and $x = 4$).

The trap is to report the local maximum only, forgetting to check the endpoint.

13 (C). There are $3^4 = 81$ strings in all. Those with no A number $2^4 = 16$, and likewise 16 have no B ; those with neither A nor B (all C) number 1. By inclusion and exclusion the strings missing an A or a B number $16 + 16 - 1 = 31$, so those with at least one of each number $81 - 31 = 50$.

14 (A). Square where useful: $2^{3/2} = 2\sqrt{2} \approx 2.83$. Compare with $e \approx 2.718$ by squaring: $(2\sqrt{2})^2 = 8 > e^2 \approx 7.39$, so $2^{3/2} > e$. The remaining values are all below e : $\ln 15 \approx 2.71$, $3^{0.9} < 3 \cdot 3^{-0.1} \approx 2.69$, $5^{0.6} \approx 2.63$ and $\pi^{0.8} \approx 2.50$. Hence $2^{3/2}$ is the largest.

15 (D). With no continuity assumed, the intermediate value theorem does not apply: a function can jump from 2 at $x = 1$ to -3 at $x = 4$ without ever taking the value 0 in between, so J does not force K and is not sufficient. Nor is it necessary: a function can have a root in $(1, 4)$ while $f(1) \neq 2$. Thus J is neither necessary nor sufficient for K . (Options E and F are false for the same reason.)

16 (C). If $\gcd(n, 6) = 1$ then n is odd and not a multiple of 3; writing $n = 6m \pm 1$ gives $n^2 = 36m^2 \pm 12m + 1 \equiv 12m(3m \pm 1) + 1$, and $m(3m \pm 1)$ is always even, so $n^2 \equiv 1 \pmod{24}$. Conversely if n shares a factor 2 or 3 with 24 the congruence fails (for instance $2^2 = 4$, $3^2 = 9$). So the condition is exactly that n is coprime to 6. (Being coprime to 6 is the same as not being a multiple of 2 or 3; “odd” alone is too weak, as $9^2 = 81 \equiv 9$.)

17 (B). To prove “ n^2 even $\Rightarrow n$ even” one may prove the logically equivalent contrapositive “ n odd $\Rightarrow n^2$ odd”, which is exactly option B. Option A proves the converse, a different statement; C only checks examples and cannot establish a universal claim; D and E prove neither.

18 (C). Draw from the box labelled “Mixed”. Its label is wrong, so it is purely Apples or purely Oranges, and the single fruit settles which. Suppose it is Apples. The box labelled “Oranges” is not Oranges (its label is wrong) and is not Apples (already located), so it must be Mixed; the last box is then Oranges. Drawing from either other box can leave two cases unresolved.

19 (D). A, B and F are the AM–GM inequality in disguise; C is $\frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$; E rearranges to the same thing. Only D can fail: at $a = b = c = 1$ it reads $3 \geq 6$, which is false. So D is the one that does not always hold.

20 (A). Add the three equations: $x^2 + y^2 + z^2 + 2(x+y+z) + 3 = 0$, that is $(x+1)^2 + (y+1)^2 + (z+1)^2 = 0$. A sum of real squares is zero only when each is zero, so $x = y = z = -1$ and $x + y + z = -3$.



tmua.fyi